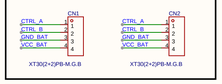
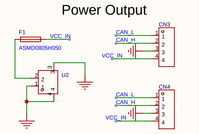
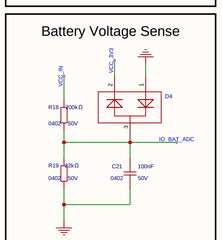
Power Input

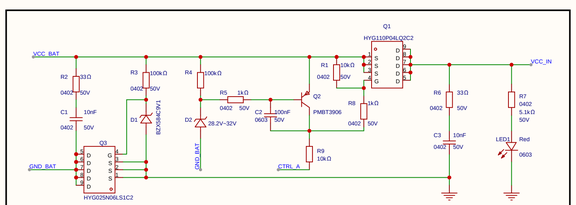


Power Output



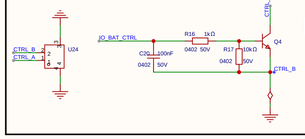
Battery Voltage Sense



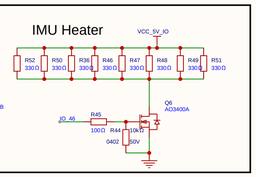


Reverse Polarity Protection, Soft Start, Overvoltage Protection
Battery Input 9 – 26V

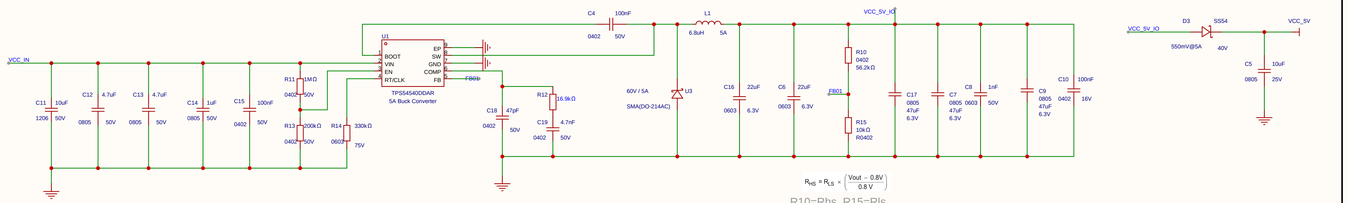
Battery Power Switch



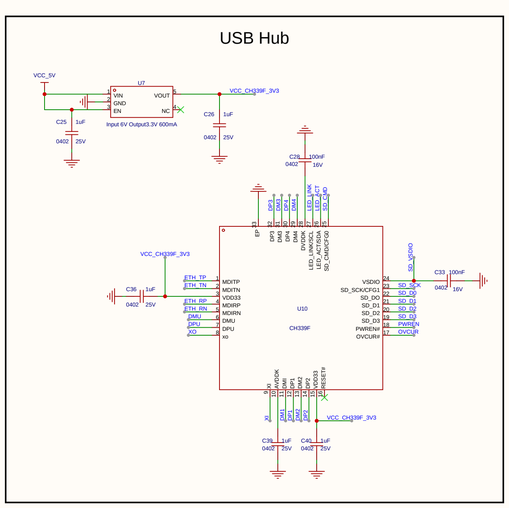
IMU Heater



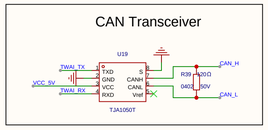
TPS54540降压电路为IO排针供电 5V5A



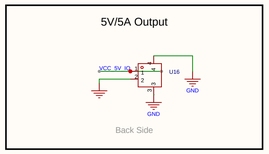
USB Hub



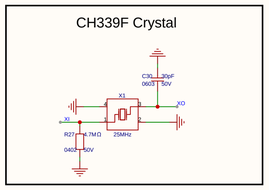
CAN Transceiver



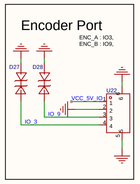
5V/5A Output



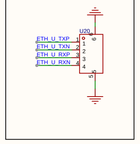
CH339F Crystal



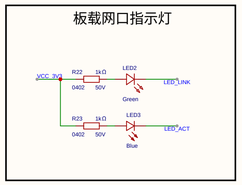
Encoder Port



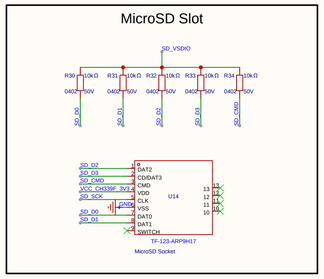
Ethernet Port



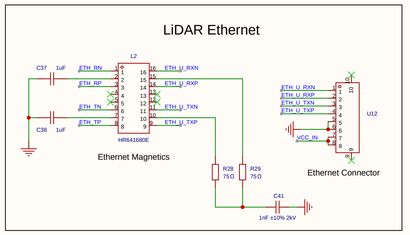
板载网口指示灯



MicroSD Slot

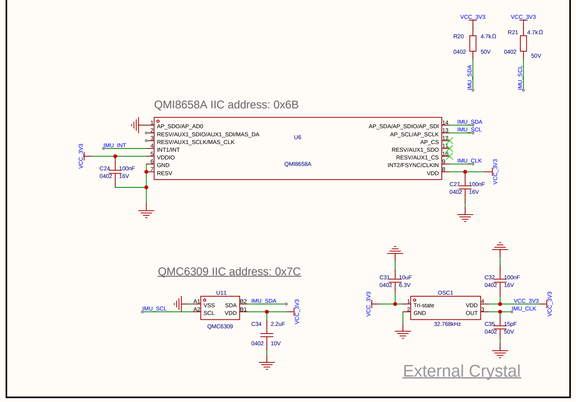


LiDAR Ethernet

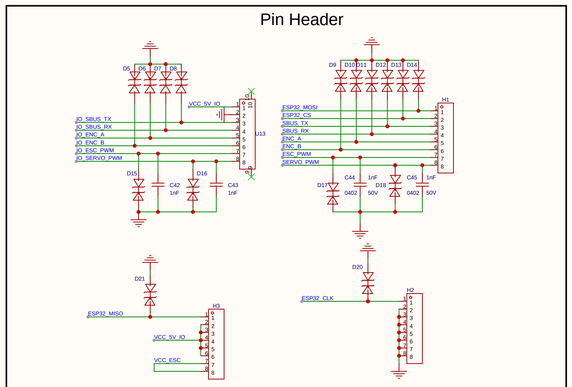


IMU Module 9-Axis IMU

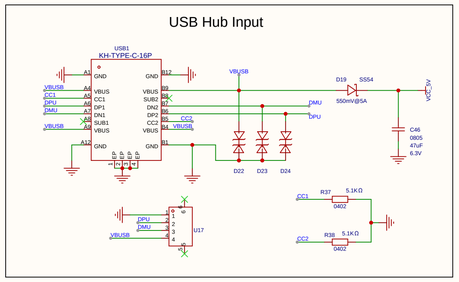
QMI8658A六轴传感器 + QMC6309三轴地磁传感器



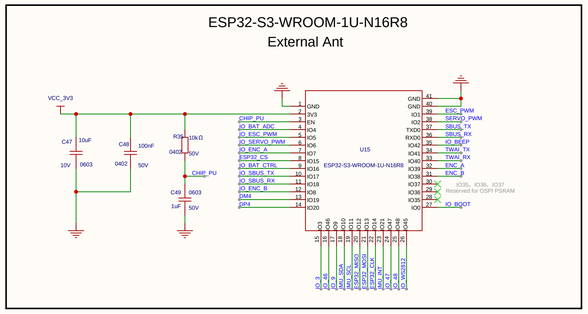
Pin Header



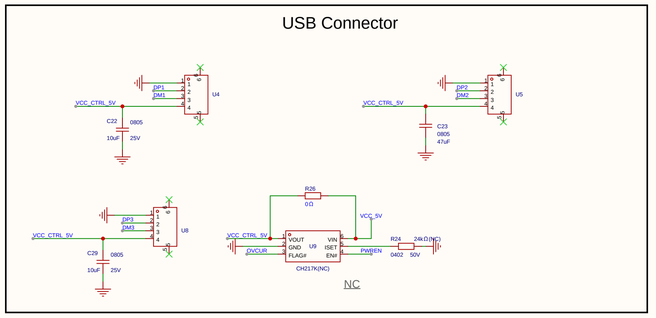
USB Hub Input



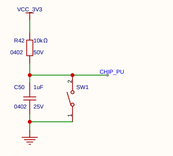
ESP32-S3-WROOM-1U-N16R8 External Ant

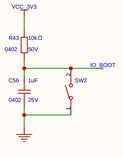


USB Connector

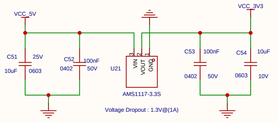


Reset / Boot

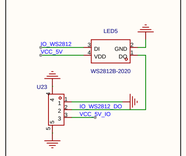




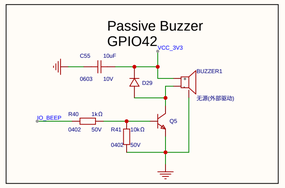
AMS1117-3.3



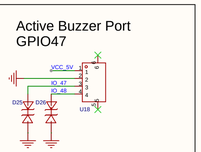
WS2812 RGB LED



Passive Buzzer GPIO42

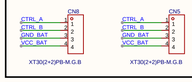


Active Buzzer Port GPIO47

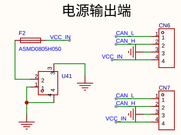


原理图		更新日期	
		创建日期	
图页		物料编码	
绘制			
审阅			
		版本	尺寸 / 页 / 共
嘉立创EDA		V1.0	OSRBOT

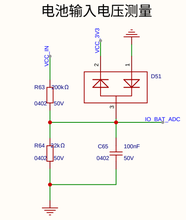
电源输入端

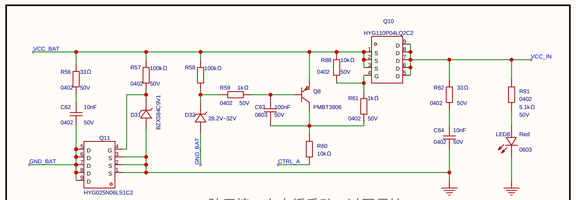


电源输出端

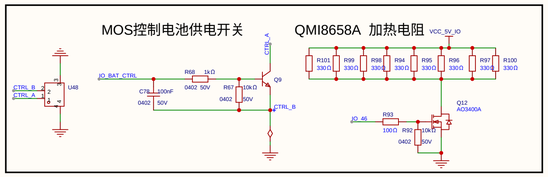


电池输入电压测量

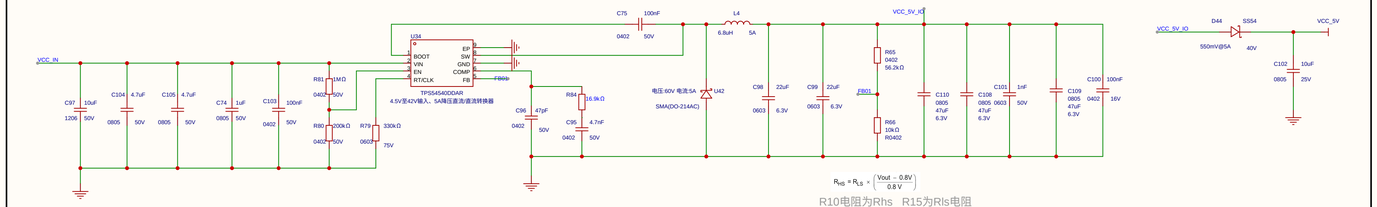




防反接，上电缓启动，过压保护
电池端输入电压9V~26V



TPS54540降压电路为IO排针供电 5V5A



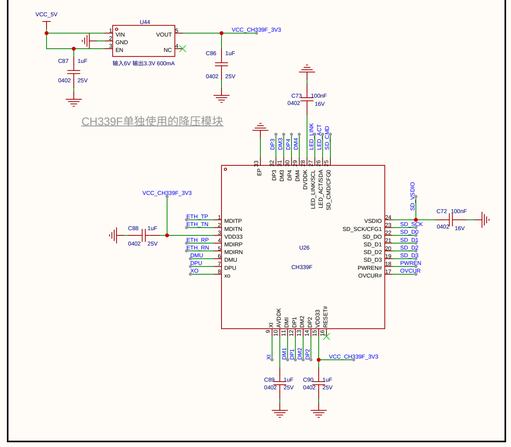
$$R_{HS} = R_{LS} \times \left(\frac{V_{out} - 0.8V}{0.8\,V} \right)$$

R10电阻为Rhs　R15为Rls电阻

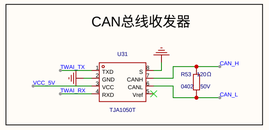
R10 : 56.2K R15:10K，输出电压5.25V(默认)
R10 : 51K R15:9.31K，输出电压5.18V

这里电容额定电压必须选择50V，为了兼容输入电池电压9-26V

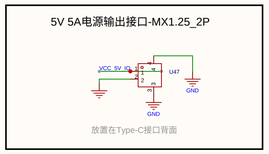
USB Hub主芯片CH339F



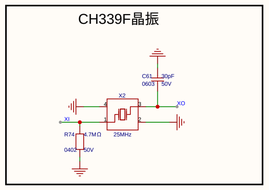
CAN总线收发器



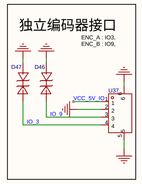
5V 5A电源输出接口-MX1.25_2P



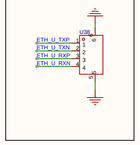
CH339F晶振



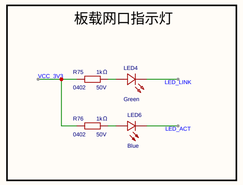
独立编码器接口



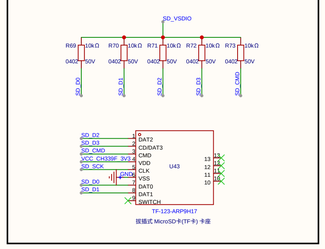
独立网络接口



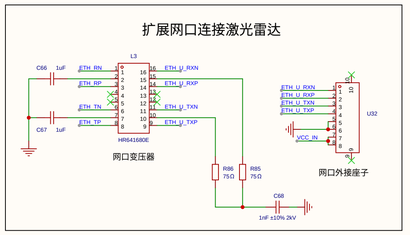
板载网口指示灯



MicroSD卡槽

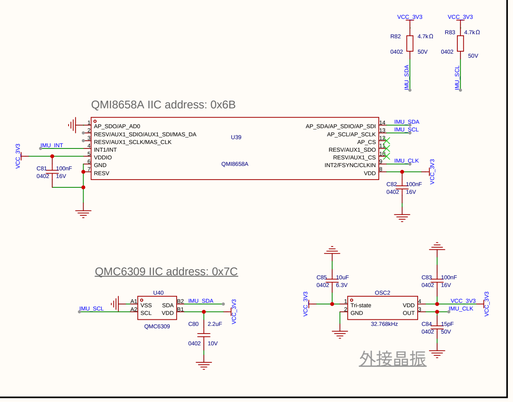


扩展网口连接激光雷达

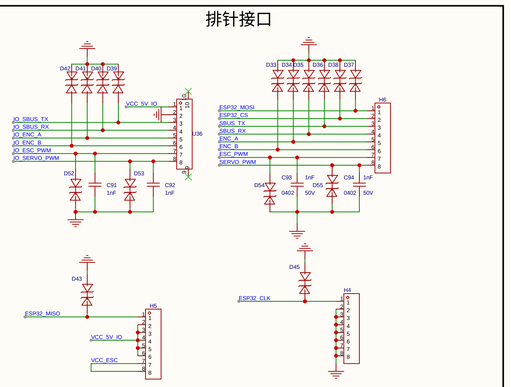


IMU模块（九轴传感器）

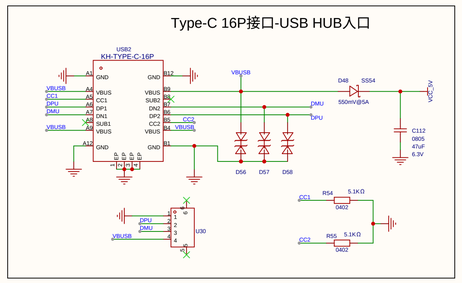
QMI8658A六轴传感器 + QMC6309三轴地磁传感器



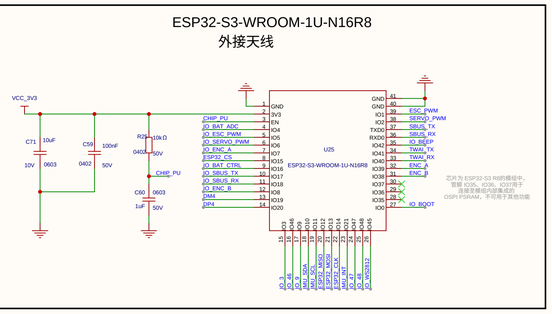
排针接口



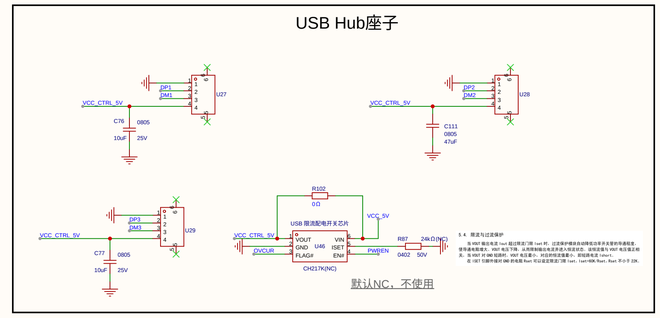
Type-C 16P接口-USB HUB入口



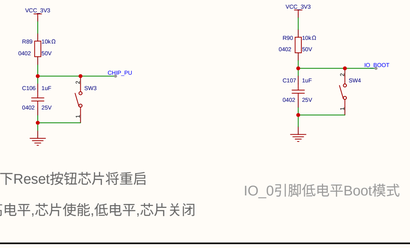
ESP32-S3-WROOM-1U-N16R8
外接天线



USB Hub座子



ESP32S3功能按钮-Reset和Boot按钮

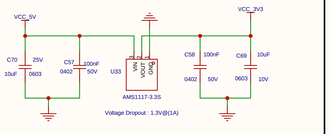


按一下Reset按钮芯片将重启

CHIP_PU高电平,芯片使能,低电平,芯片关闭

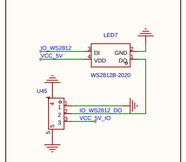
IO_0引脚低电平Boot模式

AMS1117-3.3降压电路5V-3.3V

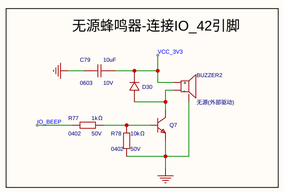


5V降压至3.3V为ESP32S3模组供电

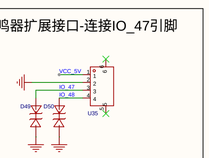
WS2812三色LED灯



无源蜂鸣器-连接IO_42引脚



有源蜂鸣器扩展接口-连接IO_47引脚



原理图		更新日期	
		创建日期	
图页		物料编码	
绘制			
审阅			
		版本	V1.0
		尺寸	
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